Calorimetry lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use an energy balance to analyze mixing and heat transfer.

SUCCESS CRITERIA - I can...

1. Identify hot and cold samples and their masses.
2. Balance energy gained and lost in an ideal model.
3. Discuss heat exchange with the container and surroundings.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “calorimetry” and “isolated” to explain your model or result. Sentence starters are optional.

calorimetry: using temperature changes to track energy transfers

isolated: exchanging no relevant quantity with surroundings in this model

heat: energy transferred because of a temperature difference

01 Warm-up: bring back an earlier idea

A 3 kg sample has c=500 J/(kg·°C) and warms by 7°C without changing phase.
Find Q.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $Q_{hot} + Q_{cold} = 0$; $Q = mc(T_f - T_i)$.
Ideal isolated mixing, no phase change. Heat lost is negative; heat gained is positive. Real surroundings may exchange energy.

Worked reasoning: Two equal masses of the same liquid at 20°C and 40°C mix in an insulated container. The final 30°C makes the hot sample lose the same energy the cool sample gains.

Your turn: Draw heat-transfer arrows and show the two equal temperature changes.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Equal masses of the same liquid at 20°C and 60°C are mixed in an ideal insulated container.
Predict final temperature and justify.

Word help: justify = support your answer with a reason or evidence

Sentence starter: I predict … . My model supports this because … .



Each outline represents the labeled sample or container.

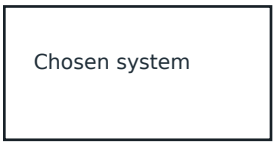
Marzano check - circle: 1 2 3

04 Apply the physics

In a mixing trial, the warm water loses 300 J. First assume only the cold water receives that energy. Then consider a cup that absorbs 50 J.
Draw both accounts and find the cold water's energy gain in each case.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

Plan a table for masses, initial temperatures, final temperature and a heat-loss observation. Why should the final temperature be measured promptly after mixing?

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A spring stores 4 J at a given compression. Within its Hooke’s-law range, find the energy at 6 times that compression.
Mark equilibrium, displacement and restoring-force direction.

Word help: compression = shortening or squeezing relative to an unstressed length

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

07 Exit check

An ideal mixture of 30°C and 70°C samples reaches 80°C with no energy added. Is that plausible?
Explain 0.

Word help: ideal = using stated simplifying conditions

Sentence starter: My claim is … . The diagram/data show …, so … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Mix one cup of 20°C water with two equal-size cups of 50°C water, neglecting losses.
Predict whether final temperature is above or below 35°C, then calculate it.

Sentence starter: I predict … . My model supports this because … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3